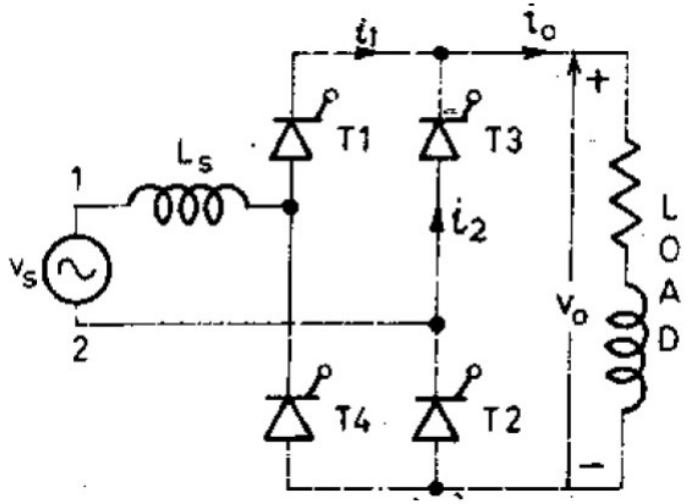
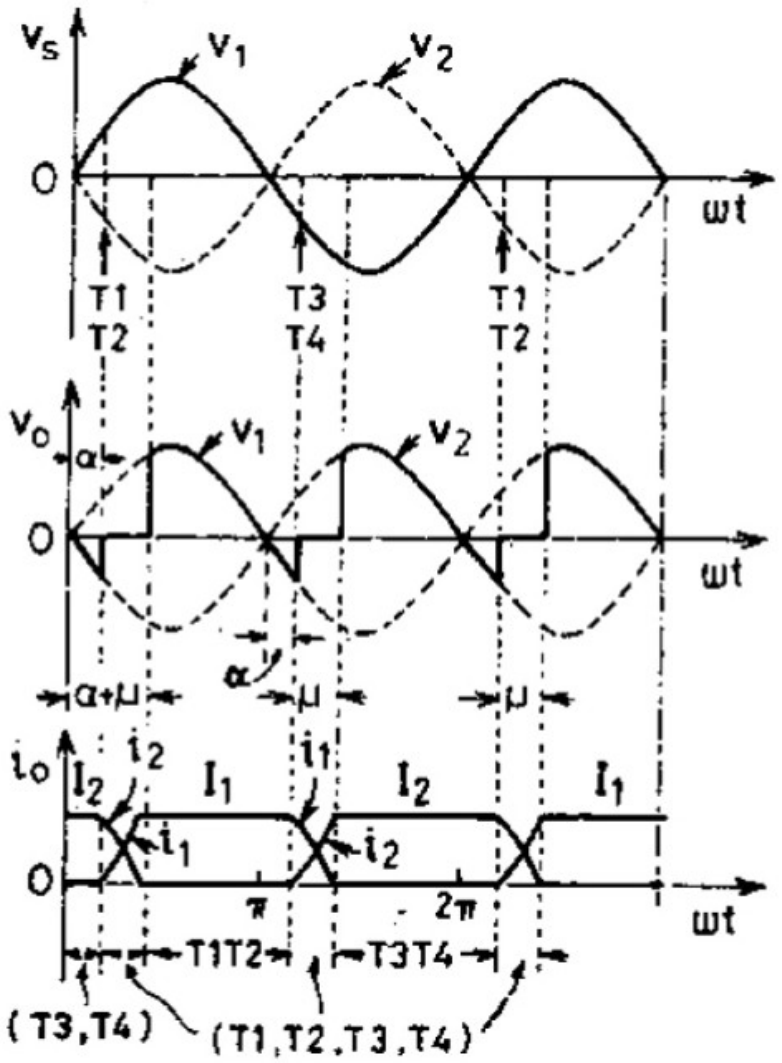
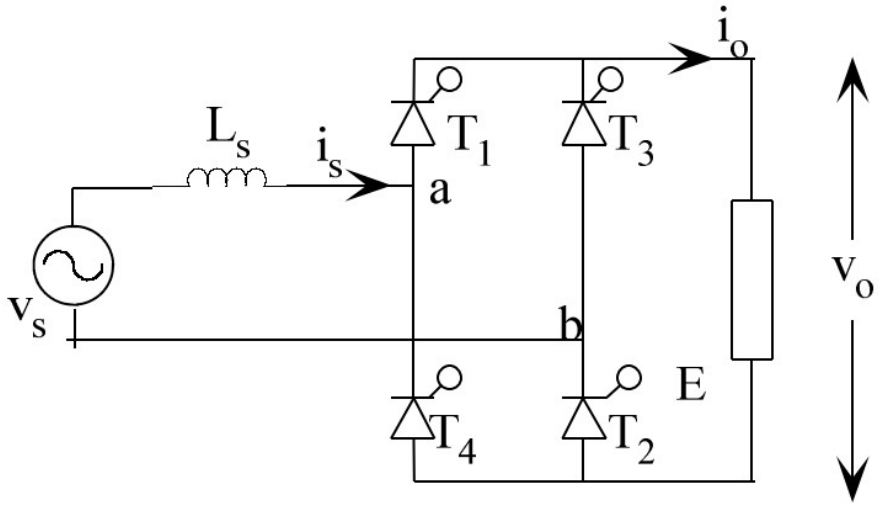
Effect of Source inductance

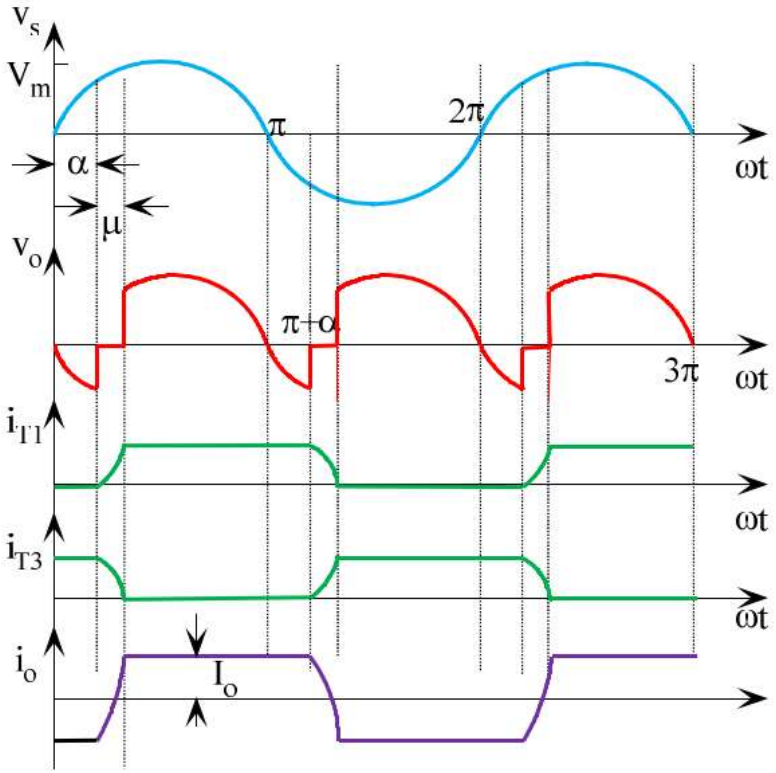




During overlap angle μ, all devices T1, T2, T3, T4 are conducting; The output is short circuited;
load voltage is zero. Input voltage appears across Ls.

Effect of Source inductance



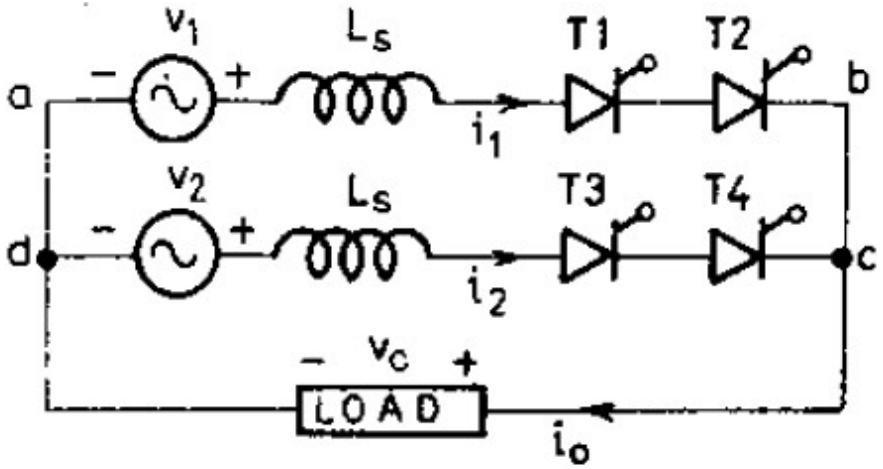


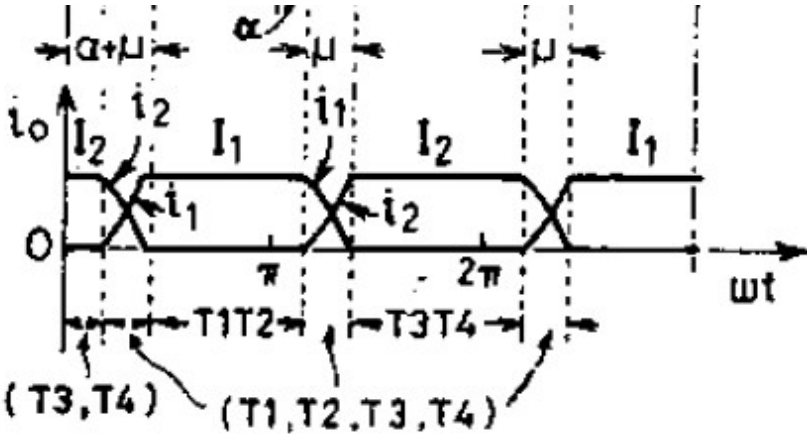
- During the positive half of the supply cycle, thyristors T1 and T2 are triggered at firing angle α.

- The positive group SCRs T1 and T2 now act as incoming SCRs
- whereas the negative group SCRs T3 and T4 will act as outgoing SCRs(previously conducting).

- Since the load current is assumed to be constant(large load inductance) ,

current should immediately transfer from outgoing SCRs to incoming SCRs, i.e., from T3 T4 to T1 T2.


v_1
L_s
T1
T2
a
b
i_1
v_2
L_s
T3
T4
d
c
i_2
v_c
LOAD
i_o



- The current, i_2 due to T3 and T4 should go to zero instantaneously and the current, i_1 due to T1 and T2 should reach the load current i_{dc} instantaneously.
- An inductor does not allow sudden changes in current.

- Due to source inductance LS, the fall of i2 and rise of i1 will not be instantaneous rather it is gradual.
- Hence, there will be a small period during which both the currents i1 and i2 will flow in the circuit,
- Resulting in a short circuit across the source terminals.
- This period is called as overlapping period.

During the overlapping period, the output voltage will be zero.

$$L_s \frac{d_{is}}{dt} = V_m \sin \omega t$$

The source current is changes from -Io to +Io during $\alpha \le \omega t \le \alpha + \mu$

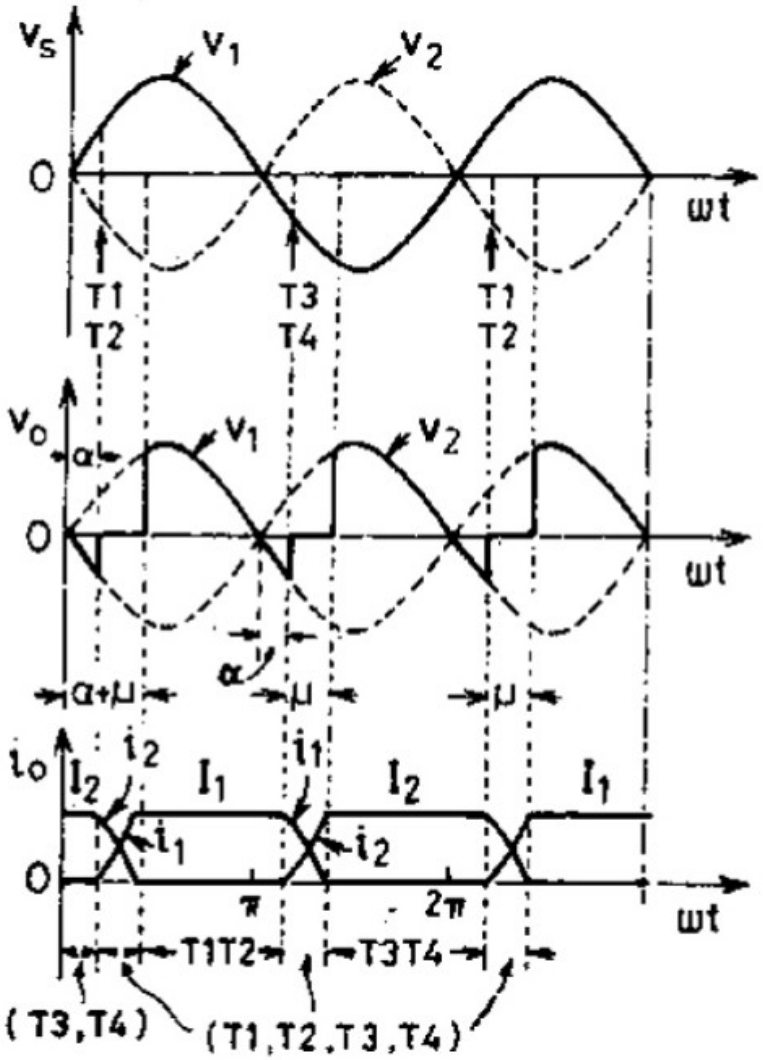
$$d_{is} = \frac{V_m}{L_s} \sin \omega t dt$$

$$\int_{-I_o}^{I_o} di_s = \frac{V_m}{\omega L_s} \int_{\alpha}^{\alpha+\mu} \sin \omega t d(\omega t) = \frac{V_m}{\omega L_s} \left[-\cos \omega t \right]_{\alpha}^{\alpha+\mu}$$

$$2I_o = \frac{V_m}{\omega L_s} \int_{\alpha}^{\alpha+\mu} \sin \omega t d(\omega t) = \frac{V_m}{\omega L_s} \left[\cos \alpha - \cos(\alpha + \mu) \right]$$

$$I_o = \frac{V_m}{2\omega L_s} \left[\cos \alpha - \cos(\alpha + \mu) \right]$$

$$\cos(\alpha + \mu) = \cos \alpha - \frac{2\omega L_s I_o}{V_m}$$

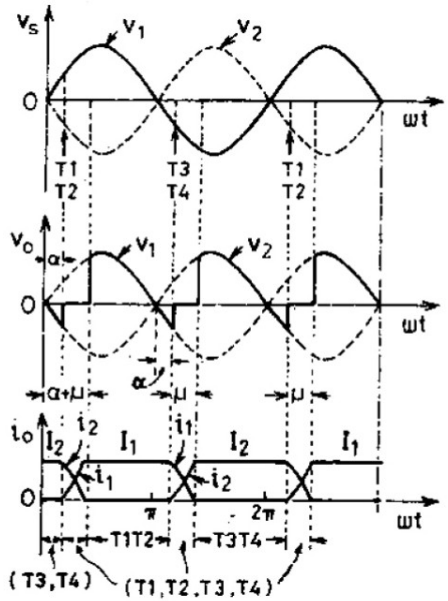


Average output voltage, $V_o = \frac{2}{2\pi}\int_{\alpha+\mu}^{\pi+\alpha} V_m \sin\omega t d(\omega t) = \frac{V_m}{\pi}\left[-\cos\omega t\right]_{\alpha+\mu}^{\pi+\alpha}$

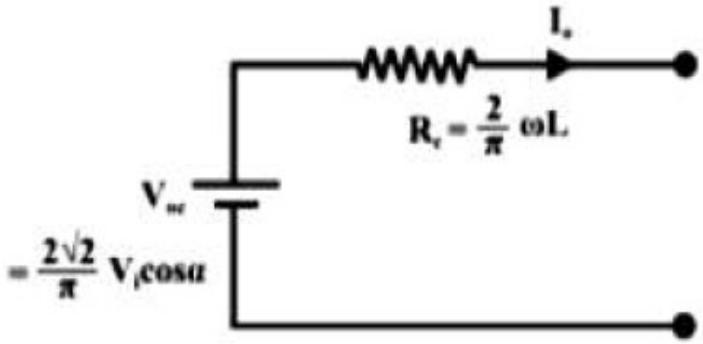
$$= \frac{V_m}{\pi}\left[\cos(\alpha+\mu)+\cos\alpha\right]$$

$$= \frac{V_m}{\pi}\left[2\cos\alpha - \frac{2\omega L_s I_o}{V_m}\right] = \frac{2V_m\cos\alpha}{\pi} - \frac{2\omega L_s I_o}{\pi}$$

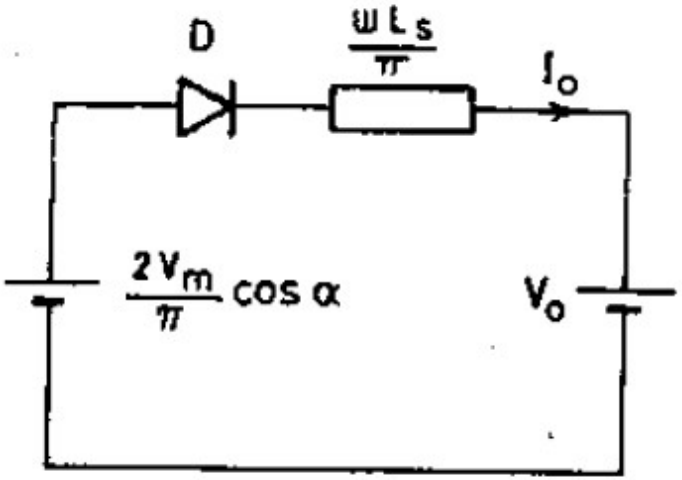
$$V_o = \frac{2V_m\cos\alpha}{\pi} - \frac{2\omega L_s I_o}{\pi}$$

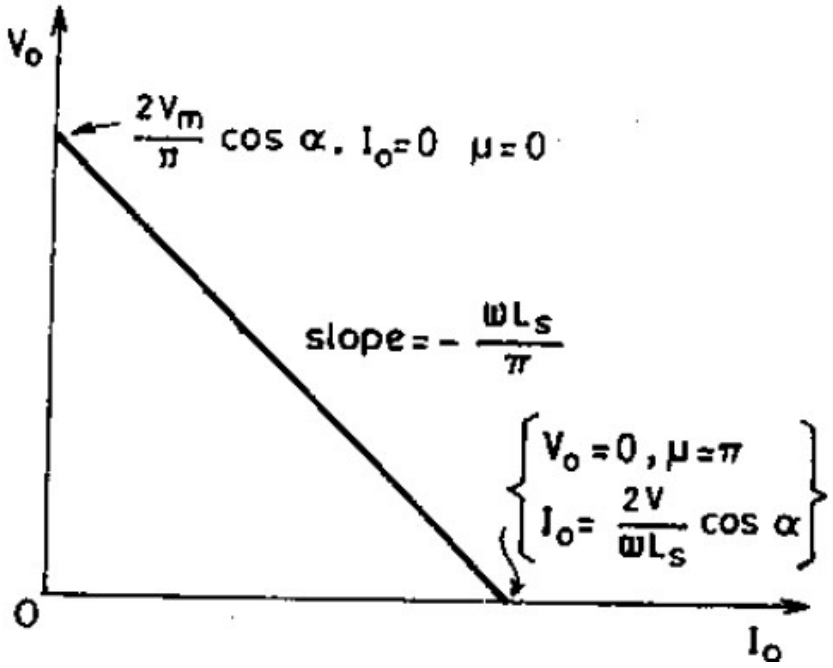


Hence, drop in average output voltage due to overlap is $\frac{2\omega L_s I_o}{\pi}$. It depends only on L_s and I_o.



Equivalent circuit


D
$\frac{\omega L_s}{\pi}$
I_o
$\frac{2V_m}{\pi}\cos\alpha$
V_o


V_o
$\frac{2V_m}{\pi}\cos\alpha,\ I_o=0\quad \mu=0$
slope $= -\frac{\omega L_s}{\pi}$
$V_o=0,\ \mu=\pi$
$I_o=\frac{2V}{\omega L_s}\cos\alpha$
O
I_o

Problem: A single-phase converter is connected to ac supply of 330sin314t volts. It operates with a firing angle $\alpha = 45°$. The total load current is maintained constant at 5A and the load voltage is 140V. Find the source inductance, angle of overlap and the load resistance.

$$V_o = \frac{2V_m \cos\alpha}{\pi} - \frac{2\omega L_s I_o}{\pi}$$

$$\cos(\alpha + \mu) = \cos\alpha - \frac{2\omega L_s I_o}{V_m}$$

$$V_o = \frac{2V_m \cos\alpha}{\pi} - \frac{2\omega L_s I_o}{\pi} = 140V$$

$L_s = 17.25mH$

$\mu = 6.315°$

$$R = \frac{V_o}{I_o} = 28\Omega$$